Notes: Agarwal and Hauswald (RFS, 2010)

Distance and Private Information in Lending

Contents

1 Big picture

- **Question.** Why does physical distance matter in small-business lending? Is it mainly because of transportation costs, or because proximity helps banks gather soft private information?
- **Setting.** The paper studies loan applications by small firms to a large U.S. bank with an extensive branch network. The key advantage of the data is that the authors observe not only credit decisions and loan terms, but also the bank's internal proprietary credit score.
- **Core challenge.** Distance can matter in at least two very different ways. It can matter directly through transportation costs and spatial price discrimination, or indirectly because nearby borrowers are easier to understand and monitor, which improves the bank's information set. Reduced-form distance effects alone do not distinguish these channels.
- **Main empirical idea.** The paper isolates the **subjective, private** component of the bank's internal score by orthogonalizing the proprietary score with publicly observable credit-quality measures. This residual becomes a proxy for soft private information.
- **Main result.** Borrower proximity improves access to credit but leads to higher borrowing costs. Once the bank's private-information measure is included, the raw distance effects on both credit availability and loan pricing largely disappear. This strongly suggests that distance works through the quality of soft information.
- **Competition result.** More distant and higher-quality borrowers are more likely to reject the bank's offer and switch lenders. This means the bank's informational advantage weakens at the geographic periphery, so competitors can poach customers more aggressively there.
- **Risk result.** Distance raises subsequent delinquency, and distance sharply reduces the predictive power of the bank's private signal for default. This is direct evidence that the bank's screen becomes less precise farther away.
- **Why the paper matters.** The paper gives unusually direct evidence that soft information is **local**, that it can be partially **hardened** inside credit-scoring systems, and that banks **strategically use** this information to soften competition in opaque credit markets.

2 Data and measurement (key objects)

2.1 Observed loan-application panel

- **Unit of observation.** A small-business loan application submitted in person at a branch of a large U.S. bank.
- **Sample period.** January 2002 to April 2003, so roughly a fifteen-month window.
- **Initial sample.** 28,761 new SME loan applications satisfying the Basel I small-business definition.
- **Final estimation sample.** After matching to credit-bureau records, geocoding addresses, and dropping incomplete observations, the main sample contains 25,487 applications.
- **Bank context.** The lender is a top-five U.S. commercial bank and the third-largest small-business lender, with 1,552 branches and market shares of roughly 30%–50% by state in its operating area.

- **Approved versus rejected applications.** The sample contains 12,823 offers and 12,664 denials.

2.2 Observed geographic variables

- **Firm-bank distance.** Driving miles, driving minutes, and aerial miles between the applicant and the lending branch.
- **Firm-competitor distance.** Driving and aerial distance to the nearest bona fide competitor branch active in small-business lending.
- **Cleaning.** The authors remove the top 1% of the distance distribution, leaving applications within 250 driving miles of the lending branch.
- **Interpretation.** These variables proxy for either transportation costs or the precision of local information, depending on the underlying model.
- **Important descriptive fact.** Only 7.21% of successful applicants go to their closest bank branch, and only 13.66% of rejected applicants do. This already suggests that small-business loans are not a homogeneous commodity and that a pure closest-bank transportation-cost story is inadequate.

2.3 Observed credit-quality measures

- **Public information.** The firm's Experian Commercial Intelliscore and the owner's personal National Risk Model score.
- **Proprietary score.** The bank's internal credit score, common in methodology across branches, ranges from 0 to 1,850 and summarizes the bank's screening outcome.
- **Soft-information component.** Branch personnel can revise the proprietary score using subjective criteria such as management quality, collateral impressions, and local business prospects.
- **Relationship proxies.** The paper measures the length of the lending relationship with *Months on Books* and the breadth of the relationship with *Scope*, a variable based on prior borrowing, deposits, and other purchased banking services.

2.4 Observed loan terms and controls

- **Loan terms.** Offered APR (all-in cost), loan amount, maturity, term-loan versus credit-line status, collateral, and personal guarantees.
- **Borrower controls.** Firm age, monthly net income, industry dummies, and local economic conditions.
- **Local controls.** ZIP-code-level Case–Shiller home price index, plus state and quarter dummies.
- **Interest-rate controls.** Maturity-matched U.S. Treasury yield and the 5-year minus 3-month Treasury term spread.

2.5 Borrower response and ex post performance

- **Switching behavior.** Among the 12,823 offers, 874 borrowers (about 6.8%) reject the bank's terms and seek credit elsewhere.
- **Matched switching outcome.** For 539 of these 874 cases (61.7%), the authors can verify that the borrower indeed accepted a competing loan.
- **Delinquency outcome.** Among booked loans, 319 become at least 60 days overdue within 18 months of origination, a delinquency rate of about 2.7%.
- **Comparison group.** Rejected applicants who obtain credit elsewhere default much more frequently, around 25%, whereas accepted applicants who later switch lenders default at about 3%.

2.6 Unobservables and timing

- **Unobservable of interest.** The quality of the bank's soft, borrower-specific private information.
- **Operational timing.** The application begins with branch-level data collection and interviews; the branch can then revise the internal score subjectively before deciding whether to offer credit and on what terms.
- **Why timing matters.** Because the authors observe all applications and all offers, not just booked loans, they can model offer decisions and quoted APRs without the usual simultaneity between accepted loan terms and borrower selection.

3 Models and empirical specifications (all equations + interpretation)

3.1 Competing theories of distance

The paper contrasts two broad classes of models.

- **Asymmetric-information view.** Distance reduces the quality of a bank's local private information. Nearby firms are easier to evaluate, which gives the bank an informational advantage over rivals. This advantage can be used strategically to soften competition and extract rents.
- **Transportation-cost view.** Distance directly affects the cost of transacting, meeting, monitoring, or shopping across branches. In that case, distance moves loan prices directly, but it should not be systematically related to the bank's private-information content or to lender switching in the same way.

This distinction leads to clear empirical predictions:

- under asymmetric information, switching to competitors should become **more likely** with firm-bank distance;
- private-information measures should be **correlated** with proximity;

- the bank should make **more screening mistakes** farther away, so delinquency should rise with distance;
- under transportation costs alone, distance affects terms directly, but it should not systematically explain information quality or ex post default once observables are controlled for.

3.2 Constructing a measure of soft private information

The proprietary score may reflect both hard and soft information, so the paper purges it of publicly observable credit-quality signals. The main regression is

$$\ln(\text{PropScore}_i) = \beta_0 + \beta_1 XCI_i + \beta_2 XNRM_i + \eta_b + u_i,$$

where:

- XCI_i is the firm's Commercial Intelliscore,
- $XNRM_i$ is the owner's National Risk Model score,
- η_b denotes branch fixed effects,
- and the residual $\hat{u}_i$ is labeled the **Subjective Credit Assessment** (SCA).

- **Interpretation.** The SCA is the component of the bank's credit assessment that is not captured by public credit scores and is therefore the paper's proxy for soft private information.
- **Why this is plausible.** The R^2 of this regression is 0.73, leaving 27% of the variation in the proprietary score unexplained, which closely matches the bank's internal claim that roughly 20%–30% of the score reflects subjective information.

3.3 Loan-offer equation

The bank's credit decision is modeled as a logistic discrete-choice problem:

$$\mathbb{P}(\text{Offer}_i = 1) = \Lambda\Big(\alpha + \delta_B \ln(1 + \text{FBDist}_i) + \delta_C \ln(1 + \text{FCDist}_i) + \theta\, \text{SCA}_i$$

$$+ \phi_B\, \text{SCA}_i \ln(1 + \text{FBDist}_i) + \phi_C\, \text{SCA}_i \ln(1 + \text{FCDist}_i) + X_i'\gamma + \eta_b\Big),$$

where $\Lambda(\cdot)$ is the logistic cdf and X_i contains relationship variables, firm controls, and local controls.

- **Baseline prediction.** If distance proxies for deteriorating information, then $\delta_B < 0$ and $\delta_C > 0$: being farther from the bank lowers approval, while being farther from the nearest competitor raises it.
- **Interaction prediction.** If proximity improves information quality, then the effect of a better SCA on approval should be weaker at larger firm-bank distances.

3.4 Loan-pricing equation

For the subsample of offers, the paper estimates an OLS model of the quoted APR:

$$\begin{aligned}\mathrm{APR}_i = {}& \alpha + \delta_B \ln(1+\mathrm{FBDist}_i) + \delta_C \ln(1+\mathrm{FCDist}_i) + \theta\,\mathrm{SCA}_i \\ & + \phi_B\,\mathrm{SCA}_i \ln(1+\mathrm{FBDist}_i) + \phi_C\,\mathrm{SCA}_i \ln(1+\mathrm{FCDist}_i) + X_i'\gamma + \eta_b + \varepsilon_i.\end{aligned}$$

- **Interpretation.** Without conditioning on private information, distance may appear to move prices directly. Once SCA is included, if distance works through information quality, the raw distance coefficients should collapse.
- **Economic logic.** When the bank trusts its local information more, it can charge more aggressively because nearby rivals face a stronger adverse-selection problem.

3.5 Borrower switching equation

To study competitor responses, the paper estimates a logistic model for whether the borrower declines the bank's offer:

$$\begin{aligned}\mathbb{P}(\mathrm{Decline}_i = 1) = \Lambda\Big(& \alpha + \delta_B \ln(1+\mathrm{FBDist}_i) + \delta_C \ln(1+\mathrm{FCDist}_i) + \theta\,\mathrm{SCA}_i \\ & + \phi_B\,\mathrm{SCA}_i \ln(1+\mathrm{FBDist}_i) + \phi_C\,\mathrm{SCA}_i \ln(1+\mathrm{FCDist}_i) + \rho\,\mathrm{APR}_i + X_i'\gamma + \eta_b\Big).\end{aligned}$$

- **Interpretation.** This is the key specification for distinguishing informational competition from transportation costs.
- **Prediction.** Under locational asymmetric information, more distant borrowers and better credit risks should switch more often because the bank's informational hold over them weakens.

3.6 Delinquency equation

For booked loans, the paper estimates a logistic model for whether the loan becomes 60 days overdue within 18 months:

$$\begin{aligned}\mathbb{P}(\mathrm{Delinq}_i = 1) = \Lambda\Big(& \alpha + \delta_B \ln(1+\mathrm{FBDist}_i) + \delta_C \ln(1+\mathrm{FCDist}_i) + \theta\,\mathrm{SCA}_i \\ & + \phi_B\,\mathrm{SCA}_i \ln(1+\mathrm{FBDist}_i) + \phi_C\,\mathrm{SCA}_i \ln(1+\mathrm{FCDist}_i) + X_i'\gamma + \eta_b\Big).\end{aligned}$$

- **Interpretation.** This is a direct test of whether the precision of the bank's information deteriorates with distance.
- **Prediction.** If the bank knows nearby firms better, then bank-borrower distance should raise delinquency in reduced form, and the inclusion of SCA should absorb that effect.

3.7 Sample-selection correction for loan pricing

Because most studies only observe booked loans, loan pricing is usually estimated on a selected sample. This paper can do more because it observes all offers, including 874 declined offers. It nevertheless also estimates a Heckman-style correction on the booked-loan sample:

$$\mathrm{APR}_i = Z_i'\beta + \lambda_i \kappa + \varepsilon_i,$$

where λ_i is the inverse Mills ratio from the prior credit-offer decision.

- **Identification.** The selection equation uses income available for a personal guarantee as the exclusion variable.
- **Interpretation.** Once SCA is included, the inverse Mills ratio becomes insignificant. This means the bank's private information is the economic object driving both selection into offers and loan pricing.

3.8 Estimation details

- All discrete-choice models are estimated by full-information maximum likelihood.
- Loan-pricing equations are estimated by OLS.
- The paper includes branch fixed effects and branch-clustered standard errors throughout.
- Results are robust when using driving minutes or aerial distances instead of driving miles.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

- A raw correlation between distance and loan approval or loan pricing is not enough. Both transportation-cost models and asymmetric-information models can produce distance gradients in offered rates.
- Relationship variables such as account scope or months on books are also not enough by themselves. They proxy for information production, but not for the realized content or quality of the bank's information.

4.2 How the paper identifies the information channel

- The central identifying step is to isolate the bank's private signal using the residual from the proprietary-score regression on public scores.
- Once SCA is included, the direct distance effects on approval and pricing become small and statistically insignificant. This is the core finding that distance operates through information quality rather than a purely direct cost channel.

4.3 Why the local-information interpretation is credible

- Public credit scores are not correlated with borrower proximity, but the bank's proprietary score is.
- In the Online Appendix, the authors show that the coefficients on distance are essentially zero in regressions for the public scores, whereas they are highly significant for the proprietary score.
- They also show that the branch-level dispersion of subjective credit assessments increases with average borrower distance, consistent with noisier soft information farther away.

4.4 Why borrower switching is such an important test

- Transportation-cost models imply that the borrower should already be at the optimal lender and therefore should not systematically switch in response to information-distance interactions.
- By contrast, the asymmetric-information model predicts more switching at the bank's periphery, because competitors can contest those borrowers more aggressively when the incumbent bank's informational advantage weakens.
- The switching regression therefore sharply discriminates between the two mechanisms.

4.5 Why delinquency is a direct test of screen quality

- If distance simply changes direct costs, there is no strong reason why it should predict ex post delinquency once controls are included.
- If distance lowers the precision of the bank's subjective assessment, then the bank should make more type-II errors farther away. The delinquency regression is therefore a direct test of the local nature of soft information.

4.6 Why the full offer sample matters

- The paper observes all applications and all offers, not just booked loans.
- This removes the usual simultaneity between loan terms and borrower acceptance that contaminates many studies of loan pricing.
- The fact that Heckman correction matters only when SCA is omitted reinforces the interpretation that private information is the relevant omitted variable.

4.7 Relation to the literature

- Relative to Petersen and Rajan (2002), the paper does not just argue that soft information is local; it measures a bank's private signal directly.
- Relative to Degryse and Ongena (2005), it goes beyond spatial price discrimination by identifying the informational content through which distance operates.
- Relative to the broader relationship-lending literature, it shows how modern scoring technologies **harden** soft information without making it nonlocal.

5 Tables and figures

5.1 Table 1: summary statistics and the "not-the-closest-bank" fact

Read:

- Successful applicants are slightly closer to the lending branch and slightly farther from competitor branches than rejected applicants.

Table 1
Summary statistics for all loan applications

	Accept			Reject			
Loan-application outcome variable	Mean	Accept median	Std Dev	Mean	Reject median	Std Dev	t-Test P-val
Loan Rate (*APR*: all-in cost of loan)	8.46%	8.12%	2.73%	N/A	N/A	N/A	N/A
Loan Amount	$46,507	$39,687	$42,755	N/A	N/A	N/A	N/A
Maturity (years)	6.68	6.14	5.39	N/A	N/A	N/A	N/A
Term Loan vs. Credit Line	22.44%		47.02%	33.73%		47.28%	0.000
Collateral	60.03%		48.30%	49.59%		49.07%	0.000
Primary Guarantor	34.03%		47.23%	38.89%		48.75%	0.000
Primary Guarantor Monthly Salary	$36,564	$33,011	$85,480	$33,378	$30,892	$92,475	0.004
SBA Guarantee	0.56%		4.70%	12.21%		27.45%	0.000
Firm-Bank Dist (miles by car)	9.91	2.62	21.44	10.67	2.98	28.94	0.017
Firm-Comp Dist (miles by car)	1.10	0.55	1.59	0.93	0.48	1.48	0.000
Firm-Bank Time (minutes by car)	10.25	6.79	21.39	13.35	7.93	22.99	0.000
Firm-Comp Time (minutes by car)	2.18	1.16	4.99	2.12	1.09	4.51	1.000
Firm-Bank Aerial Distance (miles)	7.68	2.00	23.55	8.49	2.41	17.23	0.002
Firm-Comp Aerial Distance (miles)	0.74	0.38	1.58	0.68	0.34	1.15	0.001
Internal Credit Score	1036.35	1042.44	139.32	810.72	846.89	128.79	0.000
Private Credit Assessment $\hat{u}_i$	0.0379	0.0112	0.7224	−0.0349	−0.0106	0.5830	0.000
Scope of Banking Relationship	35.14%		44.03%	25.38%		43.52%	0.000
Months on Books	43.17	30.50	56.68	17.66	14.38	29.74	0.000
Monthly Deposit Account Balance	$16,983	$11,834	$62,777	$11,549	$10,035	$21,047	0.000
Months in Business	115.39	96.34	107.28	90.88	81.03	99.28	0.000
Firm's Monthly Net Income	$110,367	$94,724	$256,941	$91,350	$84,441	$375,803	0.000
Case–Shiller HPI	170.35	156.35	30.97	162.33	150.75	31.89	0.000
Maturity-Matched UST Yield	3.89%	3.83%	1.96%	N/A	N/A	N/A	N/A
5Y–3M UST Yield Spread (bpts)	218.92	209.24	57.65	N/A	N/A	N/A	N/A
Number of observations		12,823			12,664		25,487

This table presents summary statistics for the key variables described in Section 2 for our full sample of 25,487 data points in function of the bank's decision to offer (12,823 observations) or to deny (12,664 observations) credit to the applicant. The last column indicates the P-values of a two-sided t-test for the equality of the variables' mean conditional on the bank's decision (wherever appropriate).

- The proprietary score is dramatically higher for accepted than rejected applications, which confirms that the bank's internal screening technology is economically meaningful.
- Only a very small share of firms apply to their closest bank branch: 7.21% among successful applicants and 13.66% among unsuccessful applicants.
- This is already hard to reconcile with a pure transportation-cost story in which firms should gravitate to the nearest branch.

5.2 Table 2 and Table 3: distance, private information, and the offer decision

Table 2
The decision to offer or deny credit

		1			2			3	
Specification variable	Coeff	P-val	Marg	Coeff.	P-val	Marg	Coeff.	P-val	Marg
Panel A: Distance and subjective credit assessment									
Constant	2.845	0.001		3.481	0.001		1.546	0.001	
ln(1+*Firm-Bank Dist*)	−0.092	0.001	−4.12%	−0.208	0.0724	−0.82%	−0.669	0.2067	−0.42%
ln(1+Firm-Comp Dist)	0.265	0.001	2.20%	0.615	0.0833	0.19%	0.061	0.1969	0.36%
Subjective Credit Assessment				0.739	0.001	13.98%	0.902	0.001	13.71%
SCA·ln(1+*Firm-Bank Dist*)							−0.001	0.001	−1.81%
SCA·ln(1+Firm-Comp Dist)							−0.008	0.001	−1.93%
Scope	0.710	0.001	1.03%	0.958	0.001	1.21%	0.596	0.0011	4.20%
ln(1+*Months on Books*)	1.567	0.001	21.57%	2.356	0.001	27.70%	1.392	0.001	10.66%
ln(1+*Months in Business*)	0.377	0.001	13.55%	0.230	0.001	10.86%	0.003	0.001	7.32%
ln(1+*Net Income*)	0.205	0.001	3.83%	0.581	0.001	5.08%	1.929	0.0489	4.20%
ln(1+*Case–Shiller HPI*)	0.007	0.001	3.25%	0.006	0.001	2.94%	0.006	0.001	2.97%
Collateral	0.394	0.001	11.02%	1.027	0.001	12.26%	0.479	0.001	9.45%
Personal Guarantee	3.093	0.001	0.27%	2.387	0.001	0.78%	1.600	0.001	2.07%
SBA Guarantee	−5.082	0.001	−3.23%	−4.310	0.001	−3.77%	−4.098	0.001	−3.63%
Term Loan	−0.947	0.029	−1.04%	−1.520	0.001	−2.43%	−3.764	0.001	−1.94%
4 Quarterly Dummies		Yes			Yes			Yes	
8 State Dummies		Yes			Yes			Yes	
38 SIC Dummies		Yes			Yes			Yes	
Number of Obs		25,487			25,487			25,487	
Pseudo-R^2		6.74%			10.17%			10.19%	

Table 2
(Continued)

		1			2			3	
Specification Variable	Coeff.	P-val	Marg	Coeff.	P-val	Marg	Coeff.	P-val	Marg
Panel B: Distance and public credit-quality signals									
Constant	2.836	0.001		2.807	0.001		2.790	0.001	
ln(1+*Firm-Bank Dist*)	−0.089	0.001	−4.02%	−0.090	0.001	−3.99%	−0.089	0.001	−4.00%
ln(1+Firm-Comp Dist)	0.266	0.001	2.20%	0.267	0.001	2.20%	0.255	0.001	2.19%
ln(1+Commercial Intelliscore)	0.825	0.001	9.80%				0.615	0.001	6.35%
ln(1+*National Risk Model*)				0.458	0.001	10.48%	0.443	0.001	8.17%
ln(1+XCI)·ln(1+*Firm-Bank Dist*)							0.091	0.001	1.13%
ln(1+XCI)·ln(1+Firm-Comp Dist)							0.072	0.001	0.99%
ln(1+NRM)·ln(1+*Firm-Bank Dist*)							0.086	0.001	0.83%
ln(1+NRM)·ln(1+Firm-Comp Dist)							0.185	0.001	0.96%
Scope	0.717	0.001	1.03%	0.707	0.001	1.02%	0.687	0.001	1.03%
ln(1+*Months on Books*)	1.548	0.001	21.55%	1.549	0.001	21.20%	1.559	0.001	21.55%
ln(1+*Months in Business*)	0.373	0.001	13.69%	0.371	0.001	13.53%	0.365	0.001	13.02%
ln(1+*Net Income*)	0.200	0.001	3.82%	0.199	0.001	3.79%	0.204	0.001	3.71%
ln(1+*Case–Shiller HPI*)	0.007	0.001	3.25%	0.007	0.001	3.28%	0.007	0.001	3.16%
Collateral	0.387	0.001	10.92%	0.381	0.001	10.98%	0.383	0.001	10.77%
Personal Guarantee	3.127	0.001	0.27%	3.071	0.001	0.27%	3.028	0.001	0.27%
SBA Guarantee	−4.976	0.001	−3.17%	−4.990	0.001	−3.20%	−4.882	0.001	−3.18%
Term Loan	−0.932	0.029	−1.03%	−0.933	0.029	−1.03%	−0.929	0.029	−1.02%
4 Quarterly Dummies		Yes			Yes			Yes	
8 State Dummies		Yes			Yes			Yes	
38 SIC Dummies		Yes			Yes			Yes	
Number of Obs		25,487			25,487			25,487	
Pseudo-R^2		9.07%			9.06%			9.10%	

The preceding table reports the results from estimating logistic discrete-choice models of the loan application's outcome by full-information maximum likelihood with branch fixed effects and clustered standard errors that are adjusted for heteroscedasticity across branch offices and correlation within. The dependent variable is the bank's decision to offer ($Y = 1$: 12,823 observations) or to deny ($Y = 0$: 12,664 observations) credit. The explanatory variables are the firm–bank and firm–competitor distances, the lender's pure private information (Panel A: Subjective credit assessment abbreviated SCA in the interaction terms), public credit-quality signals (Panel B: XCI is the firm's *Commercial Intelliscore* and NRM is its owner's *National Risk Model score*) bank–borrower relationship characteristics, firm attributes, and various control variables (see Section 2 for a detailed description of the variables).

We tabulate the coefficients ("Coeff"), their P-values ("P-val"), and marginal effects ("Marg") for the decision to grant credit ($Y = 1$), which we obtain by simply evaluating $\frac{\partial \Pr}{\partial x_j} = \Lambda'(\mathbf{x}_i'\beta)\beta_j$ at the regressors' sample means and coefficient estimates $\hat{\beta}$. Since the probabilities of offering and denying credit sum to 1, the marginal effects for the decision to reject a loan application are simply the opposite of the reported ones. The pseudo-R^2 is McFadden's likelihood ratio index $1 - \frac{\log L}{\log L_0}$.

Table 3
Proprietary information in the decision to offer credit

		1			2			3	
Specification variable	Coeff	P-val	Marg	Coeff.	P-val	Marg	Coeff.	P-val	Marg
Constant	2.845	0.001		3.956	0.001		1.576	0.001	
ln(1+*Firm-Bank Dist*)	−0.092	0.001	−4.12%	−0.223	0.048	−1.28%	−0.742	0.149	−0.48%
ln(1+Firm-Comp Dist)	0.265	0.001	2.20%	0.625	0.105	0.20%	0.064	0.178	0.39%
ln(1+*Proprietary Score*)				1.327	0.001	39.97%	5.925	0.001	36.86%
ln(1+Score)·ln(1+*Firm-Bank Dist*)							−0.001	0.001	−1.55%
ln(1+Score)·ln(1+Firm-Comp Dist)							−0.008	0.001	−1.57%
Scope	0.710	0.001	1.03%	1.034	0.001	1.20%	0.640	0.001	4.69%
ln(1+*Months on Books*)	1.567	0.001	21.57%	2.612	0.001	27.74%	1.404	0.001	11.05%
ln(1+*Months in Business*)	0.377	0.001	13.55%	0.241	0.001	11.64%	0.003	0.001	7.85%
ln(1+*Net Income*)	0.205	0.001	3.83%	0.599	0.001	5.30%	1.932	0.049	4.70%
ln(1+*Case–Shiller HPI*)	0.007	0.001	3.25%	0.007	0.001	3.21%	0.007	0.001	3.26%
Collateral	0.394	0.001	11.02%	1.046	0.001	13.76%	0.488	0.001	10.53%
Personal Guarantee	3.093	0.001	0.27%	2.665	0.001	0.85%	1.827	0.001	2.30%
SBA Guarantee	−5.082	0.001	−3.23%	−4.735	0.001	−3.79%	−4.627	0.001	−4.13%
Term Loan	−0.947	0.029	−1.04%	−1.583	0.001	−2.71%	−4.036	0.001	−2.21%
4 Quarterly Dummies		Yes			Yes			Yes	
8 State Dummies		Yes			Yes			Yes	
38 SIC Dummies		Yes			Yes			Yes	
Number of Obs		25,487			25,487			25,487	
Pseudo-R^2		6.74%			9.02%			9.10%	

In the above table, we replace the bank's *Subjective Credit Assessment* with its *Proprietary Score*. See the notes to table 2 for further details and Section 2 for a description of the variables.

Read:

- In the baseline specification, approval falls with firm-bank distance and rises with competitor distance. The reported marginal effects are economically meaningful: the baseline firm-bank distance effect is about −4.12%, while the competitor-distance effect is about +2.20%.
- Once the Subjective Credit Assessment is added, the distance variables lose most of their significance, which means distance works through information quality.
- Interactions show that a one-standard-deviation increase in SCA raises approval much more for nearby firms than for firms at the sample's average distance, so the bank trusts and exploits its private information more when the borrower is local.
- Replacing SCA with the broader proprietary score yields the same qualitative result, so the conclusion is not driven by the specific orthogonalization step.

5.3 Table 4 and Table 5: the pricing of credit and the role of selection

Read:

Table 4
Determinants of offered loan rates

Specification variable	1 Coeff	1 *P*-val	2 Coeff.	2 *P*-val	3 Coeff.	3 *P*-val	4 Coeff.	4 *P*-val	5 Coeff.	5 *P*-val
Constant	10.413	0.001	10.477	0.001	9.188	0.001	11.859	0.001	10.365	0.001
ln(1+*Firm-Bank Dist*)	−1.435	0.001	−0.980	0.190	−1.275	0.307	−1.010	0.187	−1.423	0.307
ln(1+Firm-Comp Dist)	1.173	0.001	0.597	0.241	1.321	0.405	0.619	0.238	1.310	0.489
Subjective Credit Assessment			−1.085	0.001	−1.127	0.001				
SCA·ln(1+*Firm-Bank Dist*)					−0.125	0.001				
SCA·ln(1+Firm-Comp Dist)					0.106	0.001				
ln(1+*Proprietary Score*)							−2.413	0.001	−2.074	0.001
ln(1+Score)·ln(1+F-B Dist)									−0.246	0.001
ln(1+Score)·ln(1+F-C Dist)									0.170	0.001
Scope	−1.516	0.001	−1.455	0.001	−1.312	0.001	−1.478	0.001	−1.329	0.001
ln(1+*Months on Books*)	−2.066	0.001	−1.871	0.001	−1.399	0.001	−2.190	0.001	−1.377	0.001
ln(1+*Months in Business*)	−3.079	0.001	−2.419	0.001	−1.432	0.001	−2.792	0.001	−1.441	0.001
ln(1+*Net Income*)	−3.514	0.001	−2.777	0.001	−1.082	0.001	−2.928	0.001	−1.105	0.001
ln(1+*Case–Shiller HPI*)	−0.479	0.001	−0.399	0.001	−0.433	0.001	−0.459	0.001	−0.461	0.001
Collateral	−2.248	0.001	−2.363	0.001	−2.337	0.001	−2.510	0.001	−2.651	0.001
Personal Guarantee	−0.880	0.001	−0.690	0.001	−1.009	0.001	−0.781	0.001	−1.159	0.001
SBA Guarantee	0.526	0.001	0.660	0.001	0.219	0.001	0.674	0.001	0.251	0.001
Term Loan	0.344	0.001	0.456	0.001	0.401	0.001	0.528	0.001	0.424	0.001
ln(1+*Maturity*)	−0.266	0.001	−0.102	0.001	−0.228	0.001	−0.119	0.001	−0.239	0.001
UST Yield	0.386	0.001	0.411	0.001	0.341	0.001	0.408	0.001	0.371	0.001
Term Spread	0.503	0.001	0.491	0.001	0.563	0.001	0.505	0.001	0.617	0.001
4 Quarterly Dummies	Yes		Yes		Yes		Yes		Yes	
8 State Dummies	Yes		Yes		Yes		Yes		Yes	
38 2-digit SIC Dummies	Yes		Yes		Yes		Yes		Yes	
Number of Obs	12,823		12,823		12,823		12,823		12,823	
Adjusted R^2	20.84%		24.26%		24.41%		22.79%		21.98%	

This table reports the results from regressing the offered loan rate (*APR*: all-in cost of the loan) on the firm–bank and firm–competitor driving distances in miles (abbreviated "F-B Dist" and "F-C Dist" in the interaction terms, respectively), soft and private information (*Subjective Credit Assessment* and *Proprietary Score*, respectively), bank–borrower relationship characteristics, firm attributes, and various control variables. We estimate the specifications by ordinary least squares with branch fixed effects and compute clustered standard errors that are adjusted for heteroscedasticity across branch offices and correlation within. See Section 2 for a description of the variables.

Table 5
Determinants of offered loan rates with sample-selectivity correction

Specification variable	1 Coeff.	1 *P*-val	2 Coeff.	2 *P*-val	3 Coeff.	3 *P*-val
Constant	10.165	0.001	11.321	0.001	11.841	0.001
ln(1+*Firm-Bank Dist*)	−1.538	0.001	−1.140	0.170	−0.711	0.665
ln(1+Firm-Comp Dist)	1.431	0.001	0.752	0.234	0.323	0.751
Subjective Credit Assessment			−1.434	0.001	−1.295	0.001
SCA·ln(1+*Firm-Bank Dist*)					−0.145	0.001
SCA·ln(1+Firm-Comp Dist)					0.074	0.001
Scope	−1.139	0.001	−1.217	0.001	−1.212	0.001
ln(1+*Months on Books*)	−4.244	0.001	−4.325	0.001	−2.160	0.001
ln(1+*Months in Business*)	−1.675	0.001	−1.578	0.001	−2.454	0.001
ln(1+*Net Income*)	−2.621	0.001	−2.597	0.001	−2.539	0.001
ln(1+*Case–Shiller HPI*)	−0.344	0.001	−0.377	0.001	−0.379	0.001
Collateral	−1.875	0.001	−1.821	0.001	−2.743	0.001
Personal Guarantee	−0.487	0.001	−0.463	0.001	−0.481	0.001
SBA Guarantee	0.206	0.001	0.199	0.024	0.797	0.001
Term Loan	0.943	0.001	0.934	0.019	1.058	0.001
ln(1+*Maturity*)	−0.042	0.001	−0.042	0.001	−0.054	0.001
UST Yield	0.455	0.001	0.517	0.001	0.736	0.001
Term Spread	0.612	0.001	0.646	0.001	0.492	0.001
Lambda	4.589	0.001	3.734	0.125	3.363	0.170
4 Quarterly Dummies	Yes		Yes		Yes	
8 State Dummies	Yes		Yes		Yes	
38 2-digit SIC Dummies	Yes		Yes		Yes	
Number of Obs	12,823		12,823		12,823	
Adjusted R^2	20.78%		23.77%		24.01%	

This table reports the results from regressing the offered loan rate (*APR*: all-in cost of the loan) on the firm–bank and firm–competitor driving distances in miles, subjective intelligence, bank–borrower relationship characteristics, firm attributes, and various control variables correcting for the bank's prior credit decision, which might result in sample-selection bias. To this end, we include Lambda, the inverse Mills ratio (hazard rate) of the Probit selection model, required by the Heckman correction. The selection equation includes the monthly income available for a personal guarantee as the identifying instrument, which we drop in the main estimation. Again, we estimate the specifications with branch fixed effects and use clustered standard errors that are adjusted for heteroscedasticity across branch offices and correlation within. Section 2 contains a detailed description of the variables.

- In the raw pricing regression, distance to the bank lowers quoted APR and distance to the competitor raises APR. Quantitatively, reducing firm-bank distance from 10 to 9 miles raises the APR by about 13 basis points, while moving the competitor from 1 to 2 miles away raises APR by about 56 basis points.
- Once SCA is included, the direct distance coefficients become insignificant. So, again, the raw spatial price pattern is really an information pattern.
- The interaction terms show that a one-standard-deviation increase in SCA lowers APR by about 103 basis points at the average firm-bank distance, but by only about 81 basis points right next to the branch. The bank prices nearby borrowers more aggressively because it has a stronger informational advantage there.
- Relationship variables continue to matter: more months on books and broader banking scope reduce offered rates substantially, consistent with the value of relationship-specific information.
- In the Heckman correction, the inverse Mills ratio matters only when private information is omitted. Once SCA is included, selection bias largely disappears, which confirms that omitted private information is the key source of bias.

5.4 Table 6: declining the loan offer and switching lenders

Read:

- Borrowers farther from the bank and closer to competitors are more likely to reject the bank's offer.
- Higher quoted APR also raises the probability of switching, which is exactly what one would

expect if competitors contest the bank's peripheral borrowers more aggressively.

- Once SCA is included, raw distance effects weaken sharply, while higher-SCA borrowers become more likely to switch. Better borrowers are the ones from whom the bank tries to extract rents, and they are also the ones competitors most want to poach.
- The interaction terms imply that bank-borrower distance weakens the incumbent bank's hold over good credit risks. This is strong evidence against transportation-cost models and in favor of locationally differentiated information.

Table 6
The decision to decline loan offers

	1			2			3		
Specification variable	Coeff	*P*-val	Marg	Coeff.	*P*-val	Marg	Coeff.	*P*-val	Marg
Constant	−2.075	0.001		−2.496	0.001		−2.385	0.001	
ln(1+*Firm-Bank Dist*)	2.587	0.001	5.50%	1.322	0.278	0.61%	0.408	0.604	1.11%
ln(1+Firm-Comp Dist)	−1.038	0.001	−7.31%	−1.280	0.384	−0.86%	−0.556	0.590	−0.05%
Subjective Credit Assessment				2.036	0.001	8.02%	1.810	0.001	10.38%
SCA·ln(1+*Firm-Bank Dist*)							0.755	0.001	1.21%
SCA·ln(1+Firm-Comp Dist)							−0.128	0.237	−0.92%
Scope	−0.971	0.001	−8.33%	−1.301	0.001	−11.81%	−1.633	0.001	−9.32%
ln(1+*Months on Books*)	−1.002	0.001	−5.90%	−1.249	0.001	−5.98%	−0.934	0.001	−5.31%
ln(1+*Months in Business*)	−4.110	0.001	−3.30%	−1.381	0.001	−4.44%	−1.017	0.001	−12.03%
ln(1+*Net Income*)	2.977	0.001	6.85%	3.688	0.001	6.56%	0.949	0.001	3.88%
ln(1+*Case–Shiller HPI*)	0.002	0.476	0.24%	0.002	0.727	0.23%	0.002	0.730	0.24%
Collateral	0.165	0.001	2.77%	0.314	0.545	4.00%	0.236	0.544	1.11%
Personal Guarantee	4.967	0.001	5.47%	5.295	0.001	6.56%	3.535	0.001	4.73%
SBA Guarantee	0.208	0.574	0.56%	0.429	0.470	0.44%	0.128	0.940	0.46%
Term Loan	−0.189	0.504	−0.17%	−0.054	0.990	−0.25%	−1.291	0.528	−0.02%
APR	0.318	0.001	11.90%	0.271	0.001	9.59%	0.708	0.001	9.86%
ln(*Loan Amount*)	−10.119	0.001	−4.14%	−7.028	0.001	−5.15%	−3.425	0.001	−3.38%
ln(1+*Maturity*)	−0.106	0.001	−2.49%	−0.120	0.001	−2.17%	−0.443	0.001	−0.88%
UST Yield	1.131	0.001	10.27%	2.218	0.001	10.40%	0.366	0.001	4.36%
Term Spread	−1.436	0.001	−3.44%	−1.722	0.001	−0.82%	−1.664	0.001	−1.31%
4 Quarterly Dummies		Yes			Yes			Yes	
8 State Dummies		Yes			Yes			Yes	
38 SIC Dummies		Yes			Yes			Yes	
Number of Obs		12,823			12,823			12,823	
Pseudo-R^2		3.27%			3.42%			3.43%	

This table reports the results from estimating a logistic discrete-choice model of the borrower's decision to refuse the bank's loan offer by full-information maximum likelihood for the subsample of successful loan applications (12,823 observations) with branch fixed effects and clustered standard errors that are adjusted for heteroscedasticity across branch offices and correlation within. The dependent variable is the applicant's decision to decline ($Y = 1$: 874 observations) or to accept ($Y = 0$: 11,949 observations) the bank's offer. The explanatory variables are the firm–bank and firm–competitor driving distances in miles, private information, bank–borrower relationship characteristics, firm attributes, and various control variables. See Section 2 for a description of the variables and the notes to table 2 for further details.

Table 7
The likelihood of credit delinquency

	1			2			3		
Specification variable	Coeff	*P*-val	Marg	Coeff.	*P*-val	Marg	Coeff.	*P*-val	Marg
Constant	−4.963	0.001		−5.659	0.001		−5.612	0.001	
ln(1+*Firm-Bank Dist*)	0.148	0.001	1.03%	0.273	0.127	0.36%	0.568	0.376	0.32%
ln(1+Firm-Comp Dist)	−0.091	0.001	−1.89%	−0.448	0.382	−0.15%	−0.499	0.207	−0.68%
Subjective Credit Assessment				−1.261	0.001	−14.35%	−1.198	0.001	−12.32%
SCA·ln(1+*Firm-Bank Dist*)							0.063	0.001	5.20%
SCA·ln(1+Firm-Comp Dist)							−0.030	0.001	−0.05%
Scope	−0.819	0.001	−8.30%	−0.543	0.001	−10.46%	−0.579	0.001	−4.45%
ln(1+*Months on Books*)	−3.025	0.001	−7.52%	−3.618	0.001	−8.88%	−3.395	0.001	−3.68%
ln(1+*Months in Business*)	−0.453	0.001	−4.13%	−0.482	0.001	−5.36%	−0.562	0.001	−5.77%
ln(1+*Net Income*)	−1.796	0.001	−1.68%	−2.663	0.001	−1.54%	−1.530	0.001	−8.37%
ln(1+*Case–Shiller HPI*)	−0.088	0.001	−5.87%	−0.081	0.001	−5.78%	−0.084	0.001	−5.81%
Collateral	−2.058	0.001	−3.33%	−3.289	0.001	−1.75%	−1.782	0.001	−3.72%
Personal Guarantee	−3.130	0.001	−6.18%	−5.154	0.001	−6.63%	−0.937	0.001	−4.59%
SBA Guarantee	6.644	0.001	4.65%	3.991	0.001	4.97%	4.914	0.001	5.00%
Term Loan	0.104	0.001	3.88%	0.478	0.001	3.83%	0.393	0.001	3.86%
APR	1.862	0.001	5.77%	2.631	0.001	7.21%	2.216	0.001	7.89%
ln(*Loan Amount*)	−3.102	0.001	−10.65%	−2.212	0.001	−11.42%	−2.395	0.001	−14.50%
ln(1+*Maturity*)	−0.201	0.623	−0.93%	−0.154	0.704	−5.03%	−0.337	0.773	−5.31%
UST Yield	0.344	0.001	2.34%	0.869	0.001	0.62%	0.206	0.001	0.47%
Term Spread	1.513	0.001	3.82%	1.639	0.001	4.17%	1.669	0.001	1.28%
4 Quarterly Dummies		Yes			Yes			Yes	
8 State Dummies		Yes			Yes			Yes	
38 SIC Dummies		Yes			Yes			Yes	
Number of Obs		11,949			11,949			11,949	
Pseudo-R^2		13.74%			17.05%			17.93%	

This table reports the results from estimating a logistic model of the likelihood that a loan becomes sixty days overdue within eighteen months of origination by full-information maximum likelihood for the subsample of actual loans booked by the bank (11,949 observations). The dependent variable is the performance status of the loan during its first eighteen months: at most sixty days overdue (corresponding to our bank's internal definition of a delinquent loan $Y = 1$: 319 observations), or current ($Y = 0$: 11,630 observations). The explanatory variables are the firm–bank and firm–competitor driving distances in miles, private information, bank–borrower relationship characteristics, firm attributes, and various control variables. See Section 2 for a description of the variables and the notes to table 2 for further details.

5.5 Table 7: delinquency and the deterioration of information quality with distance

Read:

- In reduced form, both bank distance and competitor distance help explain which booked loans later become delinquent.
- The farther away the borrower is from the bank, the more likely delinquency becomes. This means the bank commits more lending mistakes at greater distance.
- Once SCA is included, the raw distance effects disappear. That is exactly what should happen if distance matters only because it erodes the quality of the bank's private information.
- The interaction terms show that the predictive power of SCA collapses with distance. Nearby subjective assessments strongly predict repayment performance, but much less so farther away. This is the cleanest evidence in the paper that soft information is local.

5.6 Online Appendix checks: public versus private information

Read:

- Replacing SCA with public credit-quality measures in the pricing regressions does **not** make the distance variables disappear.
- Regressions of the public scores on distance produce no meaningful relation, but the proprietary score does vary with distance.
- Branches serving more distant borrowers display greater dispersion in subjective credit assessments.

- These appendix results are important because they rule out the interpretation that SCA is just a noisy proxy for omitted hard information.

6 Short summary

- **Conclusion.** Physical distance matters in small-business lending mainly because it affects the bank's ability to collect and use soft private information, not because of direct transportation costs alone.
- **Identification insight.** The paper isolates the subjective part of the bank's proprietary credit score and shows that once this private-information proxy is included, raw distance effects on approval, pricing, switching, and delinquency largely vanish.
- **Empirical lesson.** Soft information is local, strategically valuable, and only partially hardenable. It allows banks to carve out local captive markets, but that informational advantage fades with distance.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that the relevant channel linking geography to credit-market outcomes is the production and use of soft private information.
- Proximity increases the availability of credit because branches know nearby firms better.
- The same proximity also increases the price of credit because the bank can use its superior local information to soften competition and extract information rents.
- Competitors contest those rents at the bank's periphery, where distance erodes screen quality and borrowers switch more often.

7.2 Economic intuition

The paper's core intuition is simple. A nearby borrower is easier to understand: the branch knows the local economy better, can interview the owner more effectively, and can interpret ambiguous information with more confidence. This improves the bank's private signal. Because rivals know the incumbent bank is better informed near its own branch, they fear adverse selection and compete less aggressively there. The incumbent can therefore both lend more and charge more.

As the borrower gets farther away, this private informational edge deteriorates. The incumbent becomes less certain about its own signal, competitors are less afraid of adverse selection, and credit markets become more competitive. The observed consequences are exactly the ones the paper documents: more switching, lower information rents, and worse ex post screening performance at greater distance.

7.3 Takeaway

This paper is one of the cleanest empirical demonstrations that soft information is both **economically important** and **geographically local**. Its major contribution is methodological as well as substantive: by extracting the subjective component of a proprietary credit score, it turns an otherwise hidden object—soft private information—into something measurable. The resulting evidence shows not only that local information matters, but also how banks use it strategically in opaque credit markets. For any work on relationship lending, screening technologies, bank competition, or the spatial organization of finance, this paper is a benchmark.